

[GC054 - Geoengineering and Weather Modification: Bridging Scientific Uncertainties for Climate Resilience](#)

[-ybkwon04@snu.ac.kr](mailto:ybkwon04@snu.ac.kr)

Title: Weather Jiu-Jitsu: Toward Controlling Atmospheric Rivers with Deep Learning Foundation Models and Adaptive Perturbation

Abstract:

Deep learning foundation models offer a new approach for modeling the evolution of complex geophysical systems. Trained on diverse Earth system datasets, these models can learn high-dimensional spatiotemporal dynamics without relying on explicit physical equations. They have already demonstrated strong forecasting performance in applications such as global weather prediction and tropical cyclone tracking. We explore the application of foundation models to forecast and modify the tracks of Atmospheric Rivers (ARs), which are among the most impactful and dynamically sensitive weather systems in the mid-latitudes.

Our Weather Jiu-Jitsu framework leverages the intrinsic chaos of atmospheric dynamics to subtly redirect or dissipate extreme weather trajectories through precisely timed, low-energy perturbations. By analyzing the sensitivity of foundation models to small perturbations in initial conditions, we identify optimal spatio-temporal windows for control. A systematic analysis of the perturbation of the environmental conditions associated with multiple cases with the evolution of atmospheric rivers was performed using the Aurora deep learning foundation model that permits a rapid computation of the propagation of the effects of perturbations. Conditions that permit a rapid amplification of small perturbations are identified as potential targets for control mechanisms to dissipate or redirect the trajectory of the atmospheric rivers to avert catastrophic consequences at landfall. Practical implications are discussed.

We invite collaboration to develop Weather Jiu Jitsu as a community research platform.